Evaluating AI Integration at XJTLU

ADS401 Students' Perspective on AI Agents at XJTLU

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Project resource

https://lionchena.github.io/ads401/

Background

Research Background

XJTLU provides AI services such as Junmou AI to support learning. However, ADS401 students' awareness, adoption, and use remain unclear.

Research Motivation & Aim

This study explores ADS401 students' use or non-use of XJTLU AI and the reasons behind their choices, to inform future improvements to these tools.

Research Questions

RQ1 **Awareness & Adoption**

To what extent are students in our ADS401 module aware of and adopting XJTLU AI?

RQ2 **Usage Patterns**

How do students in ADS401 use XJTLU AI for their academic tasks?

RQ3 **Adoption Factors & Perceived Comparison**

What perceived benefits, barriers, and differences between XJTLU AI and external AI services are associated with students' adoption and intentions to continue, try, or resume using XJTLU AI?

Perceived usefulness, perceived ease of use, and user acceptance of information technology
Davis, F. D. (1989). MIS Quarterly
To adopt or to ban? Student perceptions and use of generative AI in higher education
Farinosi, M., & Melchior, C. (2025). Humanities and Social Sciences Communications

Research Methodology

Explanatory Sequential Mixed-Methods Design

Target Participants

50 enrolled → **4** researchers excluded → **46** potentially eligible invitees

Voluntary survey participation

Achieved sample to be reported after collection.

> *The survey identifies how students use different AI services, while follow-up interviews explore why they make those choices.*

Using mixed-methods sequential explanatory design: From theory to practice
Ivankova, N. V., Creswell, J. W., & Stick, S. L. (2006). Field Methods

Survey Design

Cross-sectional online questionnaire targeting ADS401 students.

Planned grouping by reported AI use for ADS401 in the past four weeks:

G1	**Both AI**
G2	**Other AI Only**
G3	No AI
G4	XJTLU AI Only

Various academic activity categories will be used for consistent comparison.

Measures:

- Awareness
- Ever-use
- Recent Academic Usage
- Perceived Usefulness
- Ease of Use
- Trust
- Barriers
- Future-use Intentions

Usefulness and ease-of-use items: exploratory, context-specific measures; not complete validated TAM scales.

TAM (Technology Acceptance Model): validated usefulness and ease-of-use scales (Davis, 1989); these constructs inform XJTLU AI items.

Student perspectives on AI in Higher Education: Student Survey
Chung, J., et al. (2024). Monash University
Perceived usefulness, perceived ease of use, and user acceptance of information technology
Davis, F. D. (1989). MIS Quarterly

Data Analysis

Descriptive Statistics:	Report counts and percentages for awareness, ever-use, and past-four-week usage, with the relevant response denominator for each item.
Cross-tabulation:	Compare usage patterns across four groups.
Rating Analysis:	Summarise activity-specific usefulness, usefulness/ease-of-use/trust items, and future-use intentions descriptively; exclude N/A from rating summaries.
Perceived Comparison:	Examine paired activity ratings only where dual users rated both tools for the same category; report direct comparison responses separately, including 'Unable to compare'.

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Follow-up Interviews & Integrated Analysis

Semi-structured Interviews

Sampling:

Purposive sampling of opt-in respondents, informed by initial survey analysis.

Participants:

About 4–6 students; prioritise Both AI / Other AI Only.

Duration:

15–20 minutes per interview.

Focus:

Tool-choice motivations, adoption barriers, and learning experiences.

Qualitative & Integrated Analysis

Thematic Analysis:

Code interviews and identify recurring themes.

Integration:

Connect interview themes with survey patterns.

Outcome:

Explain adoption decisions and identify potential XJTLU AI improvements.

Using thematic analysis in psychology
Braun, V., & Clarke, V. (2006). Qualitative Research in Psychology
Achieving integration in mixed methods designs—Principles and practices
Fetters, M. D., Curry, L. A., & Creswell, J. W. (2013). Health Services Research

Next Steps

01 Research Materials

02 Internal Questionnaire Check (4 researchers)

03 Survey Data Collection

04 Preliminary Survey Analysis

05 Follow-up Interviews

06 Thematic Analysis & Integration

Limitations

Single-module sample

- Findings describe participating ADS401 students; no generalisation to all XJTLU students.
- Voluntary participation may limit representativeness even within ADS401.

Self-reported data

The study relies mainly on students' reported behaviors and perceptions.

Interview coverage

Prioritising Both AI and Other AI Only may underrepresent No AI and XJTLU AI Only perspectives.

Cross-sectional design

The survey captures recent use and future intentions, not causal effects or observed long-term continued use.

Perception-based comparison

Comparisons with external AI tools do not establish objective differences in system capabilities.

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Materials

Our online materials are available on:
https://github.com/LionChenA/ads401/tree/main/assets/pre2

- Study scope, RQs, sampling and analysis plan
- Full questionnaire
- Survey routing source
- Survey routing image
- Reference list

Thank you

Q&A

AI usage:
[1] GLM-5.3 Flash (Zhipu AI) and Gemini 3.8 Flash (Google), accessed via the omp agent harness. Used for slide layout and formatting, grammatical proofreading, and reference formatting.
[2] ChatGPT 6 Astra (OpenAI), available at https://chatgpt.com. Used to organise research materials, locate relevant literature, and generate and revise slides based on the supplied research script.